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NOTES

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Geometric Properties of the Gamma Function

Patrick Ahern and Walter Rudin

The fact that $\log \Gamma(x)$ is convex on the positive real axis is one of the crucial properties of Γ. Combined with the functional equation $\Gamma(x+1) = x\Gamma(x)$ and the normalization $\Gamma(1) = 1$, it uniquely characterizes the gamma function. This theorem, due to Bohr and Mollerup, is the basis of Artin's elegant treatment of $\Gamma(x)$ in [**2**].

We extend this convexity property into the complex plane, use this extension to obtain information about the argument of $\Gamma(z)$ on vertical lines, and describe some features of the conformal mappings induced by Γ'/Γ and by $\log \Gamma$. In spite of the apparently inexhaustible supply of formulas, identities, expansions, and integrals concerned with Γ, we have found no mention of such geometric properties in the literature.

For brevity, we use the notation

$$G(z) = \log \Gamma(z).$$

We let Ω_a denote the right half-plane consisting of all complex z having $\operatorname{Re} z > a$, and denote its closure by $\overline{\Omega}_a$.

Theorem 1. (A) *If* $x \geq \frac{1}{2}$ *then* $\operatorname{Re} G''(x+iy) > 0$ *for all real* y.

(B) *If* $x < \frac{1}{2}$ *then* $\operatorname{Re} G''(x+iy) < 0$ *for all sufficiently large* y.

Note that $\operatorname{Re} G''(x+iy) = (\partial^2/\partial x^2) \log |\Gamma(x+iy)|$. The theorem asserts therefore *that* $\log |\Gamma(x+iy)|$ *is a convex function of* x *in* $\Omega_{1/2}$, *but in no larger half-plane.*

Proof: We begin with (B) since its proof gave us a hint that $1/2$ might be the cut-off between (A) and (B).

Define $\psi(s) = \frac{1}{2} - s$ on $(0, 1]$, $\psi(s+1) = \psi(s)$ for all real s. One way of writing Stirling's formula is [**4**; p. 151]

$$\log \Gamma(z) = \left(z - \tfrac{1}{2}\right) \log z - z + \tfrac{1}{2} \log 2\pi + \int_0^\infty \frac{\psi(s)}{s+z}\, dx \tag{1}$$

for all $z \neq 0$ that are not negative real numbers. Differentiate this twice, then perform an integration by parts. The result is

$$G''(z) = \frac{1}{z} + \frac{1}{2z^2} + 6\int_0^\infty \frac{\varphi(s)}{(s+z)^4}\, ds, \tag{2}$$

where $\varphi(s) = \int_0^s \psi(t)\, dt$. Since ψ has mean-value 0, φ is periodic, hence bounded.

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In fact, $0 \le \varphi(s) \le 1/8$. It follows that

$$\operatorname{Re} G''(x+iy) \le \frac{2x^3+x^2+(2x-1)y^2}{2(x^2+y^2)^2} + \frac{3}{4}\int_0^\infty \frac{ds}{\left[(s+x)^2+y^2\right]^2}.$$

The last integral is $O(y^{-3})$, as $y \nearrow \infty$. When $2x-1 \neq 0$, the dominant term, for fixed x and large y, is thus $(x-\frac{1}{2})y^{-2}$, and this is negative when $x < 1/2$; (B) follows.

Next, we apply log to the identity [**2**; p. 24]

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)},$$

then differentiate twice, and obtain

$$G''(z) + G''(1-z) = \frac{\pi^2}{\sin^2(\pi z)}.$$

Since $\sin \pi(\frac{1}{2}+iy) = \cos(\pi iy) = \cosh(\pi y)$, it follows that

$$G''\left(\tfrac{1}{2}+iy\right) + G''\left(\tfrac{1}{2}-iy\right) = \frac{\pi^2}{\cosh^2(\pi y)}$$

or

$$2\operatorname{Re} G''\left(\tfrac{1}{2}+iy\right) = \frac{\pi^2}{\cosh^2(\pi y)} > 0. \tag{3}$$

Finally, (2) shows that G'' is bounded in Ω_δ, for every $\delta > 0$. Since bounded harmonic functions in a half-plane are the Poisson integrals of their boundary values, (A) follows from (3).

The following monotonicity property was needed in [**1**], but only when $b-a$ is an integer, and in that case a more direct proof exists.

Theorem 2. (i) *If* $\frac{1}{2} \le a < b$ *then*

$$\arg \frac{\Gamma(b+iy)}{\Gamma(a+iy)}$$

is an increasing function of y *on* $(-\infty, \infty)$.

(ii) *The same conclusion holds if* $0 < a < \frac{1}{2}$ *and* $b > 1-a$.

Proof: Let u and v be the real and imaginary parts of $G = \log \Gamma$, respectively. Then $v = \arg \Gamma$, the Cauchy-Riemann equations give $u_x = v_y$, and hence $v_{xy} = u_{xx} > 0$ in $\overline{\Omega}_{1/2}$, by Theorem 1. This means that $v_y(a+iy) < v_y(b+iy)$, or

$$\frac{\partial}{\partial y} \arg \Gamma(a+iy) < \frac{\partial}{\partial y} \arg \Gamma(b+iy).$$

This proves (i).

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To deduce (ii), note that

$$\frac{\Gamma(b+iy)}{\Gamma(a+iy)} = \frac{\Gamma(b+iy)}{\Gamma(1-a+iy)} \cdot \frac{\Gamma(1-a+iy)}{\Gamma(a+iy)} \cdot \frac{\Gamma(a-iy)}{\Gamma(a-iy)}$$
$$= \frac{1}{|\Gamma(a+iy)|^2} \cdot \frac{\Gamma(b+iy)}{\Gamma(1-a+iy)} \cdot \frac{\pi}{\sin \pi(a-iy)}. \tag{4}$$

Since now $1-a > 1/2$, (ii) follows from (i) and the fact that the argument of the last factor in (4) is

$$\arctan[\cot(\pi a)\tanh(\pi y)],$$

which is an increasing function of y when $0 < a < 1/2$.

The proofs of the next two theorems will use the following sufficiency criterion for univalence.

Suppose: (a) Π *is an open half-plane that does not contain the origin*, (b) f *is holomorphic in a convex region* Ω, *and* (c) $f'(z) \in \Pi$ *for every* $z \in \Omega$. *Then* f *is univalent in* Ω.

When $\Pi = \Omega_0$ this is proved on p. 47 of [**3**]. It is clear that the criterion is rotation-invariant.

Theorem 3. (i) Γ'/Γ *is univalent in* Ω_0, *but in no larger half-plane.*

(ii) $\operatorname{Re}(\Gamma'/\Gamma)(x+iy) \geq (\Gamma'/\Gamma)(x)$ *in* Ω_0.

(iii) $\operatorname{Im}(\Gamma'/\Gamma)$ *is bounded in* Ω_δ, *for every* $\delta > 0$.

(iv) $|\operatorname{Im}(\Gamma'/\Gamma)(z)| < \pi/2$ *in* $\overline{\Omega}_{1/2}$.

Proof: Since $(\Gamma'/\Gamma)' = G''$, it follows from Theorem 1 and the sufficiency criterion that Γ'/Γ is univalent in $\Omega_{1/2}$. This proves only part of (i), but at least it points in the right direction.

Differentiation of $(\Gamma'/\Gamma)(x+iy) = (u+iv)(x+iy)$ (where u and v are now the real and imaginary parts of Γ'/Γ) with respect to y gives

$$iG''(x+iy) = (u_y + iv_y)(x+iy).$$

Thus $v_y = \operatorname{Re} G''$, hence (3) becomes

$$v_y\left(\tfrac{1}{2}+iy\right) = \frac{\pi^2}{2\cosh^2(\pi y)}.$$

If we integrate this we obtain

$$v\left(\tfrac{1}{2}+iy\right) = \frac{\pi}{2}\tanh(\pi y),$$

and this implies (iv).

Differentiation of the logarithm of

$$\frac{1}{\Gamma(z)} = ze^{\gamma z}\prod_1^\infty \left(1+\frac{z}{n}\right)e^{-z/n}$$

[**4**; p. 150], where γ is Euler's constant, leads to

$$u(x+iy) = -\gamma - \frac{x}{x^2+y^2} + \sum_1^\infty \left(\frac{1}{n} - \frac{n+x}{(n+x)^2+y^2}\right), \tag{5}$$

$$v(x+iy) = \sum_0^\infty \frac{y}{(n+x)^2+y^2}, \tag{6}$$

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and

$$G''(z) = \sum_0^\infty \frac{1}{(n+z)^2}. \tag{7}$$

The right side of (5) is an increasing function of y^2, if $x > 0$. For fixed x it therefore attains its minimum when $y = 0$. This proves (ii), and (iii) follows from (6), because

$$\begin{aligned} v(x+iy) &< \frac{y}{x^2+y^2} + \sum_1^\infty \frac{y}{n^2+y^2} \\ &< \frac{1}{2x} + \int_0^\infty \frac{y}{t^2+y^2}\,dt = \frac{1}{2x} + \frac{\pi}{2}. \end{aligned}$$

For the proof of (i), let $\Pi^+ = \{y > 0\}$ and $\Pi^- = \{y < 0\}$ be the upper and lower half-planes, respectively.

If $z \in \Omega_0 \cap \Pi^+$ then $(n+z)^2 \in \Pi^+$ for all $n \geq 0$, hence $(n+z)^{-2} \in \Pi^-$, so that $G''(z) \in \Pi^-$.

Likewise, $G''(z) \in \Pi^+$ if $z \in \Omega_0 \cap \Pi^-$.

It follows that Γ'/Γ is univalent in each of the quadrants $\Omega_0 \cap \Pi^+$ and $\Omega_0 \Pi^-$. By (6), Γ'/Γ maps $\Omega_0 \cap \Pi^+$ into Π^+ and $\Omega_0 \cap \Pi^-$ into Π^-; since Γ'/Γ is strictly increasing on the positive real axis, we conclude that Γ'/Γ is univalent in all of Ω_0.

Finally, Γ'/Γ has a pole at $z = 0$, hence maps each neighborhood of 0 onto a neighborhood of ∞. Since $(\Gamma'/\Gamma)(x) \nearrow \infty$ as $x \nearrow \infty$, it follows that Γ'/Γ is not univalent in Ω_δ if $\delta < 0$.

In the next theorem, x_0 is the unique positive number at which $\Gamma'(x_0) = 0$. Since $\Gamma(1) = \Gamma(2)$, $1 < x_0 < 2$.

Theorem 4. $\log \Gamma$ *is univalent in* Ω_{x_0}, *but in no larger half-plane.*

Proof: Recall that $(\log \Gamma)' = \Gamma'/\Gamma$. If $x > x_0$, it follows from part (ii) of Theorem 3 that

$$\operatorname{Re}(\log \Gamma)'(x+iy) \geq (\Gamma'/\Gamma)(x) > (\Gamma'/\Gamma)(x_0) = 0.$$

Thus $\log \Gamma$ is univalent in Ω_{x_0}. Clearly, x_0 has no neighborhood in which $\log \Gamma$ is univalent, since $(\log \Gamma)'(x_0) = 0$.

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